Supplementary Appendix

This appendix has been provided by the authors to give readers additional information about their work.

Supplement to: Yao-Sheng Shang, Peng-Yu Zhong, Ying Ma, Nan Bai, Ying Niu, Zhi-Lu Wang. Efficacy and Safety of Proton Pump Inhibitors in Patients with Coronary Artery Diseases Receiving Oral Antiplatelet Agents and/or Anticoagulants：A Systematic Review and Meta-Analysis.

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Supplementary Table S1. Exploratory meta-regression analyses for MACCE and GI events.

Outcomes	Potential influencing factors	*P* value > | t |
MACCE		
	PPIs types	0.591
	different races	0.929
	antithrombotic strategies	0.468
	risk of bias of included trials	0.715
	publication years	0.123
	follow-up durations	0.656
	sample sizes	0.873
GI events		
	PPIs types	0.895
	different races	0.037
	antithrombotic strategies	0.000
	risk of bias of included trials	0.669
	publication years	0.089
	follow-up durations	0.519
	sample sizes	0.058

Supplementary Table S2. Sensitivity analysis for MACCE and GI events on patients after percutaneous coronary intervention.

Outcomes	Incidence rate ratio	95% confidence interval	Pr > Chi 2
MACCE	1.06	0.69-1.62	0.994
GI events	0.21	0.14-0.32	0.349

Supplementary Table S3. GRADE evidence quality of outcomes summary.

Proton Pump Inhibitor in addition to Antithrombotic strategy compared to Antithrombotic strategy for Coronary Artery Diseases

Patient or population: Coronary Artery Diseases
Settings:
Intervention: Proton Pump Inhibitor in addition to Antithrombotic strategy
Comparison: Antithrombotic strategy

Outcomes	Illustrative comparative risks* (95% CI)		Relative effect (95% CI)	No of Participants (studies)	Quality of the evidence (GRADE)	Comments
	Assumed risk **Antithrombotic strategy**	Corresponding risk **Proton Pump Inhibitor in addition to Antithrombotic strategy**				
Major adverse cardiovascular and cerebrovascular events Follow-up: 1 to 24 months	**Study population**		**RR 1.05** (0.96 to 1.15)	25277 (15 studies)	⊕⊕⊕⊖ **moderate**[2]	
	92 per 1000[1]	**96 per 1000** (88 to 106)[1]				
	Moderate					
	1	1				
All-cause death Follow-up: 3.5 to 24 months	**Study population**		**RR 0.84** (0.69 to 1.01)	22120 (7 studies)	⊕⊕⊕⊖ **moderate**[2]	
	27 per 1000	**23 per 1000** (19 to 27)				
	Moderate					
Gastrointestinal events Follow-up: 1 to 36 months	**Study population**		**RR 0.34** (0.21 to 0.54)	26059 (14 studies)	⊕⊖⊖⊖ **very low**[2,3,4]	
	25 per 1000	**9 per 1000** (5 to 14)				
	Moderate					
Cardiovascular death Follow-up: 1 to 24 months	**Study population**		**RR 0.88** (0.69 to 1.12)	19596 (5 studies)	⊕⊕⊕⊖ **moderate**[2]	
	20 per 1000	**17 per 1000** (14 to 22)				
	Moderate					
Myocardial infarction Follow-up: 1 to 24 months	**Study population**		**RR 0.98** (0.88 to 1.09)	22575 (11 studies)	⊕⊕⊕⊖ **moderate**[2]	
	70 per 1000	**69 per 1000** (62 to 76)				
	Moderate					
Stent thrombosis (ARC definite or probable) Follow-up: 1 to 24 months	**Study population**		**RR 1.01** (0.76 to 1.34)	14874 (3 studies)	⊕⊕⊕⊖ **moderate**[2]	
	19 per 1000	**19 per 1000** (14 to 25)				
	Moderate					
Gastroduodenal ulcer Follow-up: 3.5 to 12 months	**Study population**		**RR 0.40** (0.13 to 1.29)	5770 (2 studies)	⊕⊕⊕⊖ **moderate**[5]	
	3 per 1000	**1 per 1000** (0 to 4)				
	Moderate					
Overt bleeding with a gastroduodenal lesion Follow-up: 3.5 to 12 months	**Study population**		**RR 0.09** (0.03 to 0.28)	4311 (4 studies)	⊕⊕⊕⊖ **moderate**[2]	
	16 per 1000	**1 per 1000** (0 to 4)				
	Moderate					

*The basis for the **assumed risk** (e.g. the median control group risk across studies) is provided in footnotes. The **corresponding risk** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: Confidence interval; **RR:** Risk ratio;

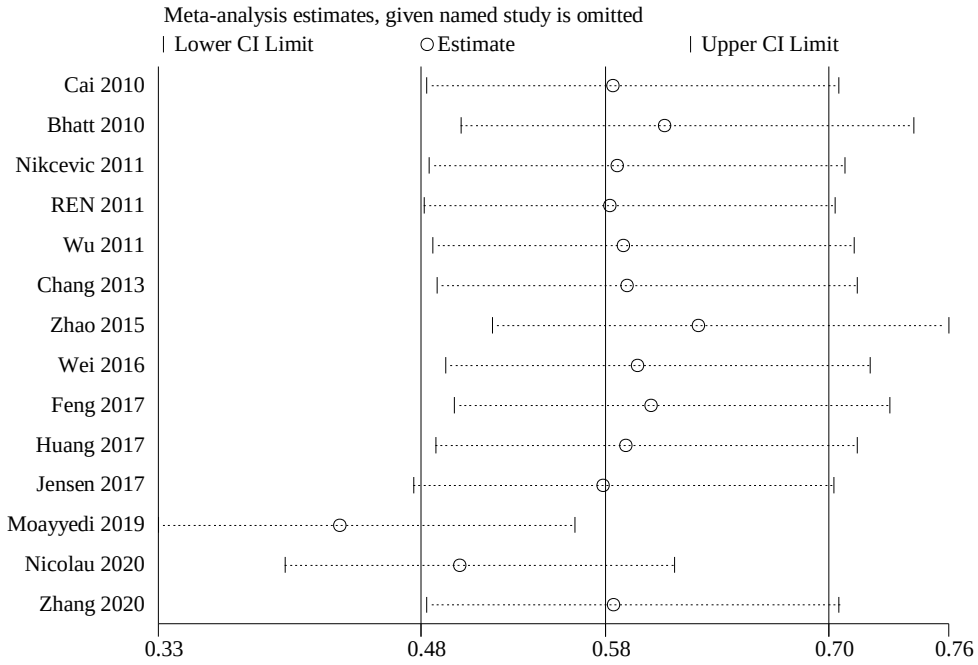
GRADE Working Group grades of evidence
High quality: Further research is very unlikely to change our confidence in the estimate of effect.
Moderate quality: Further research is likely to have an important impact on our confidence in the estimate of effect and may change the estimate.
Low quality: Further research is very likely to have an important impact on our confidence in the estimate of effect and is likely to change the estimate.
Very low quality: We are very uncertain about the estimate.

[1] Wang 2013 trial was not included in the GRADE system because it did not have the specific number of events, only the effect size.
[2] Not all trials were double-blind.
[3] The study existed significant heterogeneity.
[4] The study existed publication bias.
[5] The confidence interval was too wide.

Supplementary Table S4. The *P* value of Begg's text and Egger's text for each outcome.

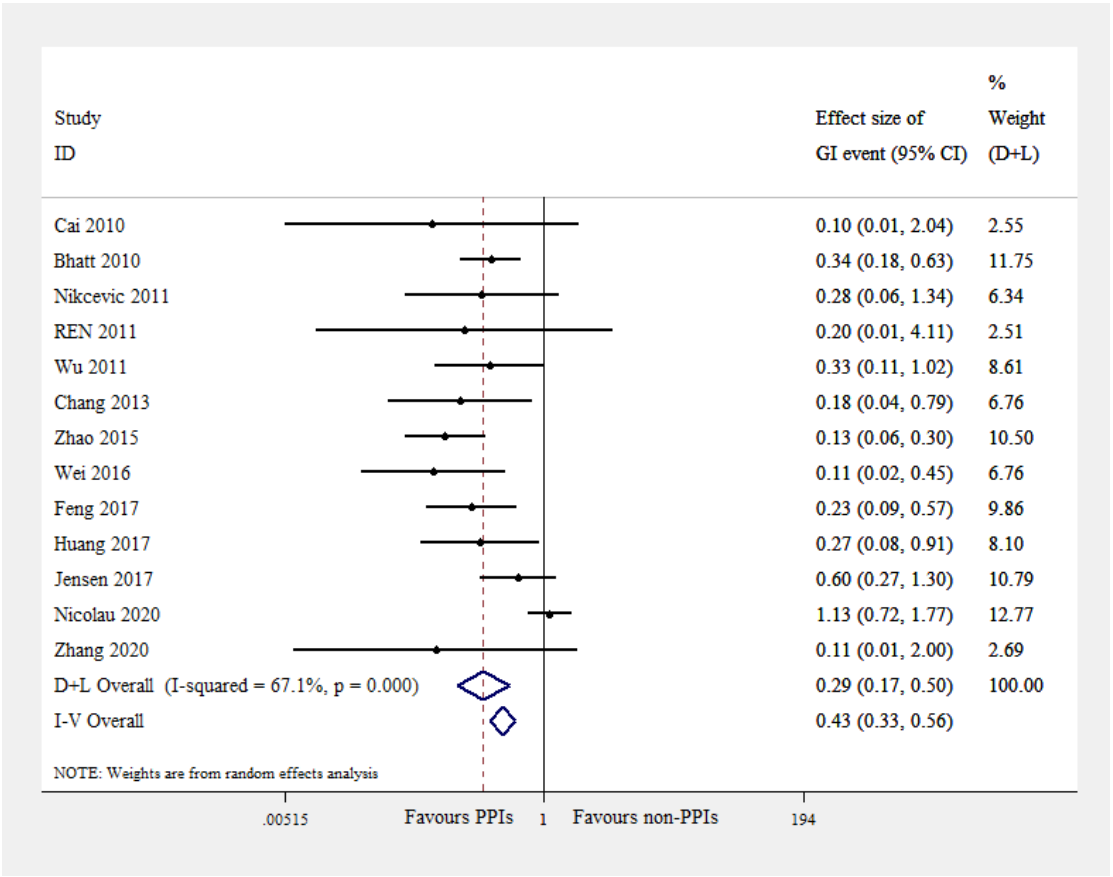
Outcomes	Pr > | z | (continuity corrected)[a]	*P* > | t |[b]
MACCE	0.902	0.625
GI events	0.743	0.002
All-cause death	0.266	0.377
Cardiovascular death	1.000	0.938
Myocardial infarction	0.373	0.386
Stent thrombosis	0.734	0.741
significant bleeding from gastroduodenal lesions	0.734	0.116
Gastroduodenal ulcer	1.000	-

Notes: a, this *P* value is derived from the Begg's text; b, this *P* value is derived from the Egger's text.



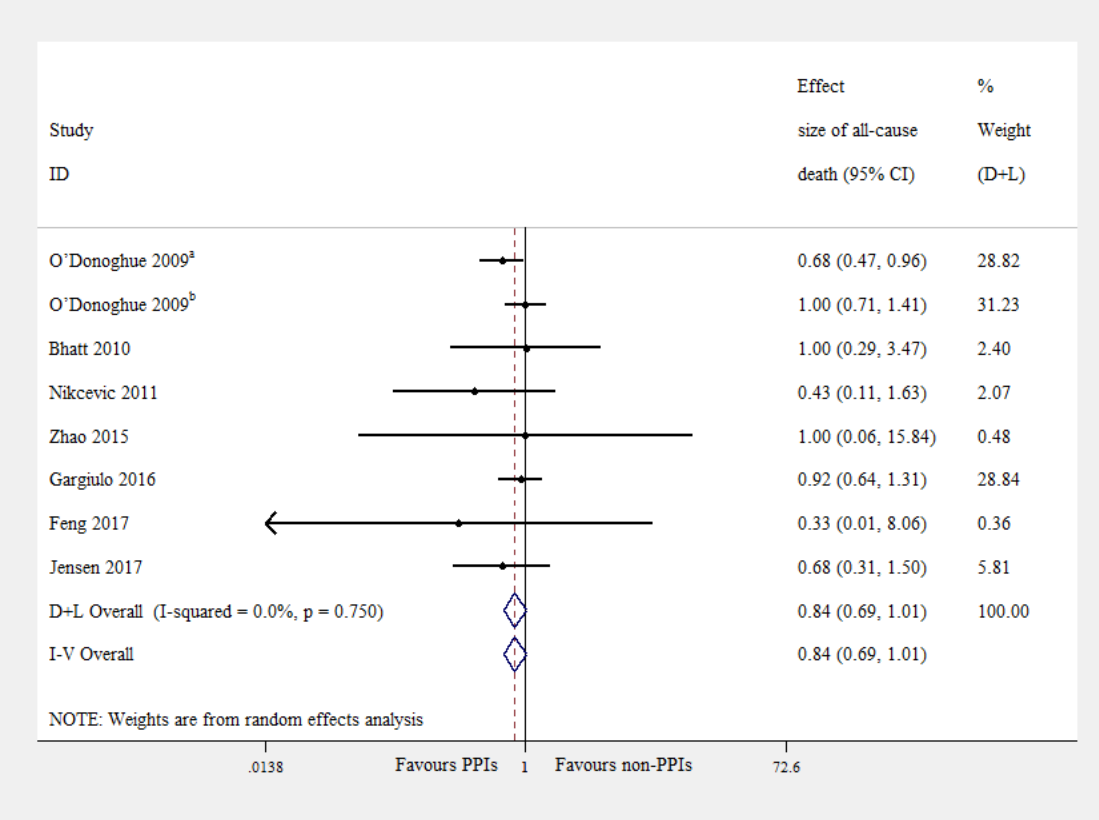
Supplementary Figure S1. "Leave-one-out" method to find the source of heterogeneity of GI events.

Notes: GI, gastrointestinal; CI, confidence interval.



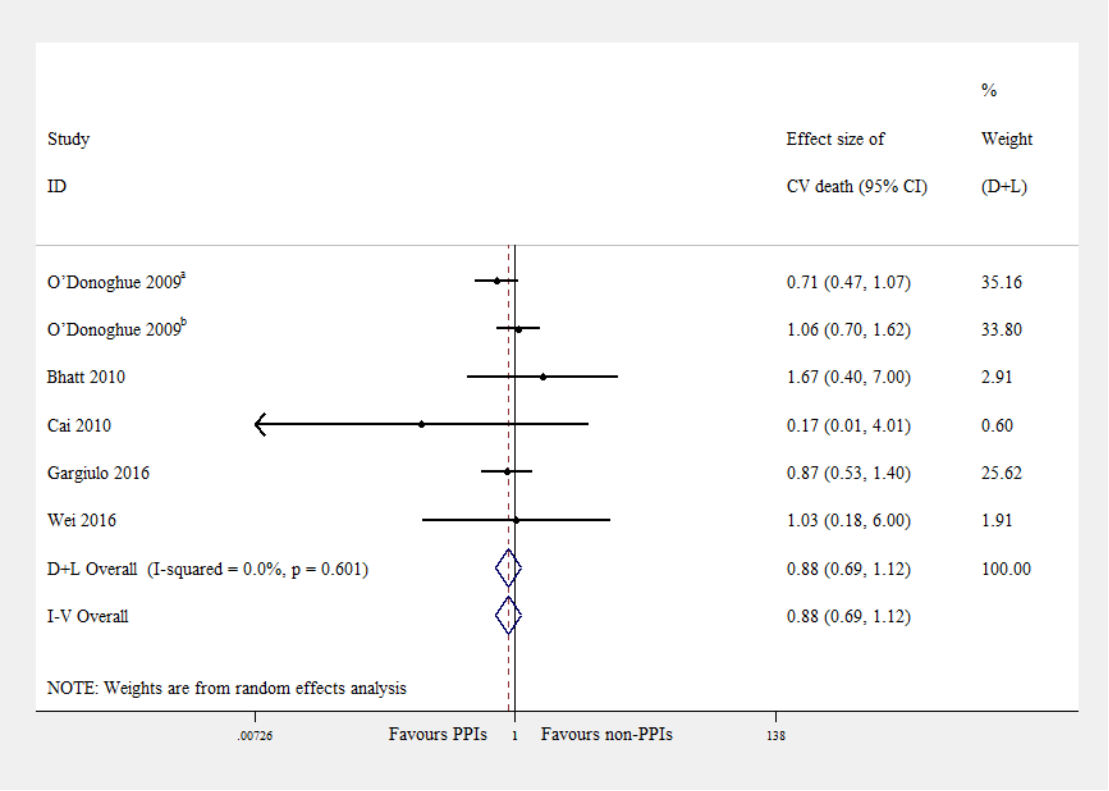
Supplementary Figure S2. Effect of PPIs combined with antithrombotic strategy on the risk of GI events in patients with coronary artery diseases after removing Moayyedi 2019 trial.

Notes: Study ID, the name of trials included; GI, gastrointestinal; D+L, the DerSimonian & Laird random effects model; M-H, the Mantel-Haenszel fixed effect model; PPIs, proton pump inhibitors



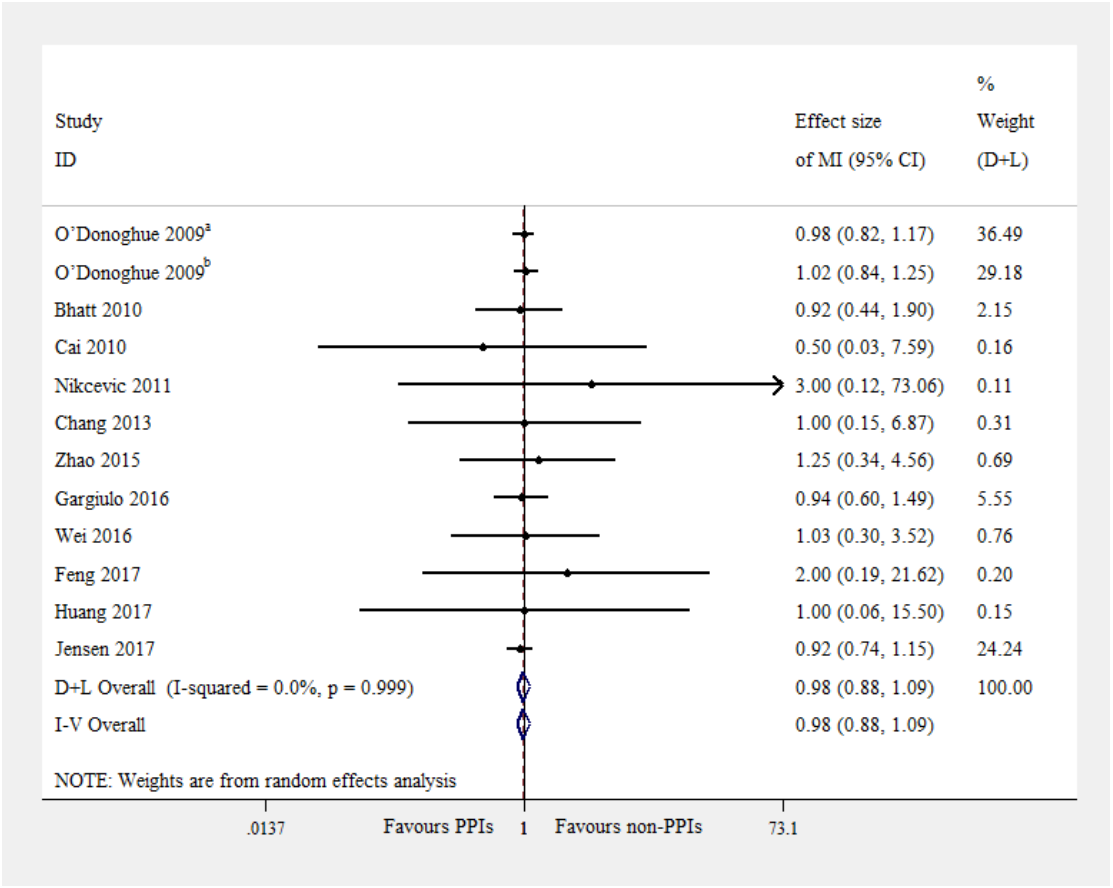
Supplementary Figure S3. Effect of PPIs combined with antithrombotic drugs on the incidence of all-cause death in patients with coronary artery diseases.

Notes: a, aspirin plus clopidogrel in the O'Donoghue trial; b, aspirin plus prasugrel in the O'Donoghue trial; D+L, the DerSimonian & Laird random effects model; M-H, the Mantel-Haenszel fixed effect model; PPIs, proton pump inhibitors.



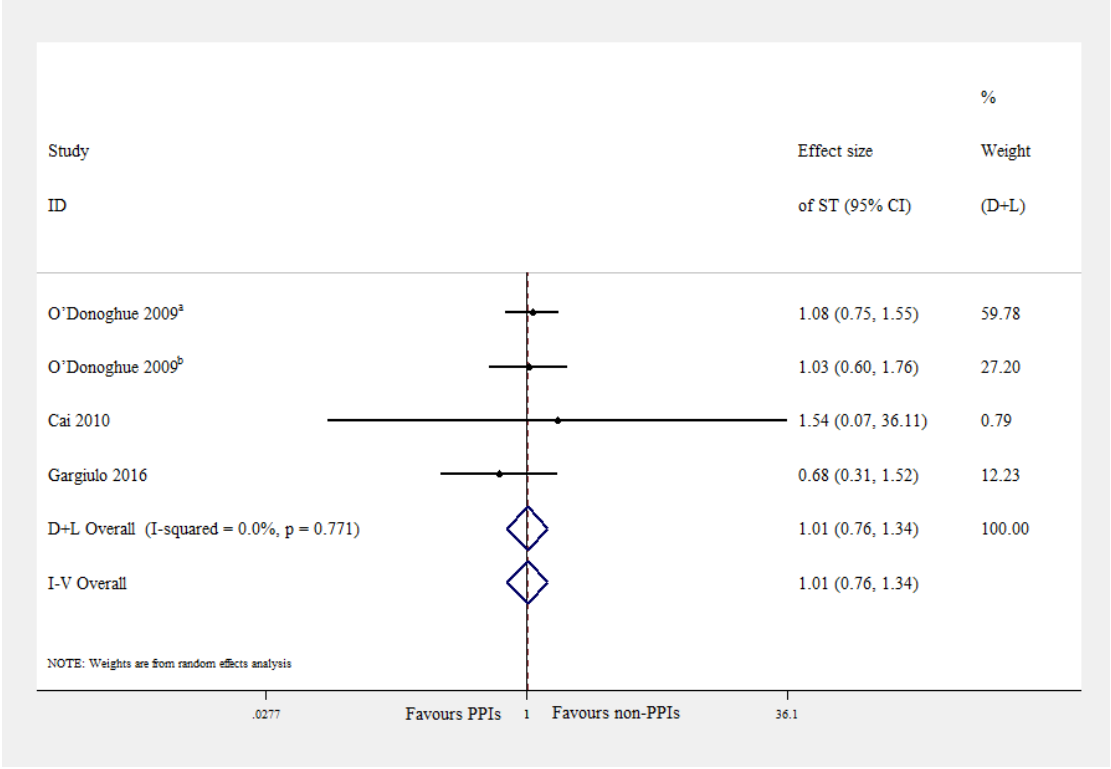
Supplementary Figure S4. Effect of PPIs combined with antithrombotic drugs on the incidence of cardiovascular death in patients with coronary artery diseases.

Notes: a, aspirin plus clopidogrel in the O'Donoghue trial; b, aspirin plus prasugrel in the O'Donoghue trial; CV death, cardiovascular death; D+L, the DerSimonian & Laird random effects model; M-H, the Mantel-Haenszel fixed effect model; PPIs, proton pump inhibitors.



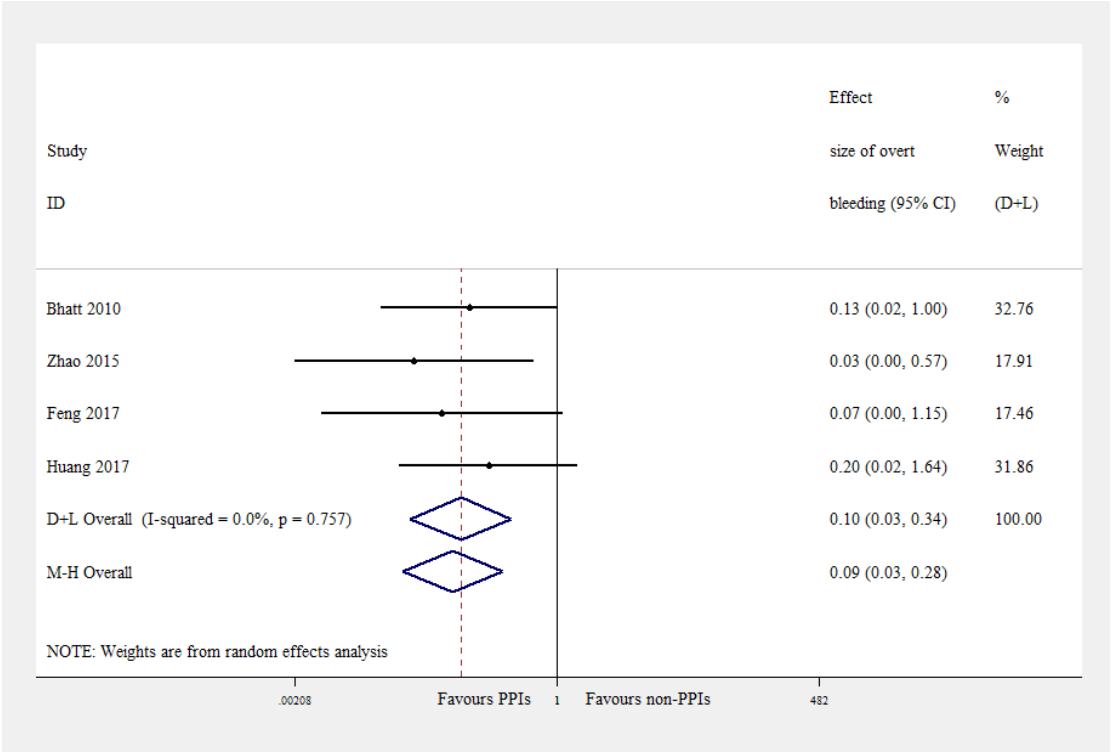
Supplementary Figure S5. Effect of PPIs combined with antithrombotic drugs on the risk of myocardial infarction in patients with coronary artery diseases.

Notes: a, aspirin plus clopidogrel in the O'Donoghue trial; b, aspirin plus prasugrel in the O'Donoghue trial; MI, myocardial infarction; D+L, the DerSimonian & Laird random effects model; M-H, the Mantel-Haenszel fixed effect model; PPIs, proton pump inhibitors.



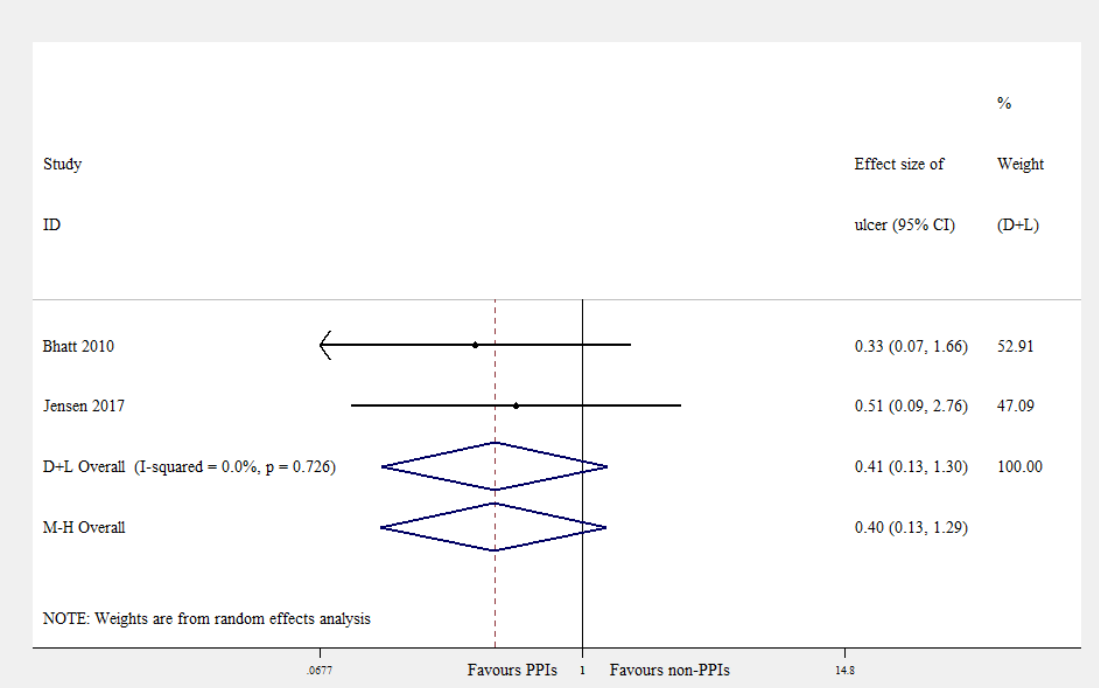
Supplementary Figure S6. Effect of PPIs combined with antithrombotic drugs on the incidence of definite or probable stent thrombosis in patients with coronary artery diseases.

Notes: a, aspirin plus clopidogrel in the O'Donoghue trial; b, aspirin plus prasugrel in the O'Donoghue trial; ST, stent thrombosis; D+L, the DerSimonian & Laird random effects model; M-H, the Mantel-Haenszel fixed effect model; PPIs, proton pump inhibitors.



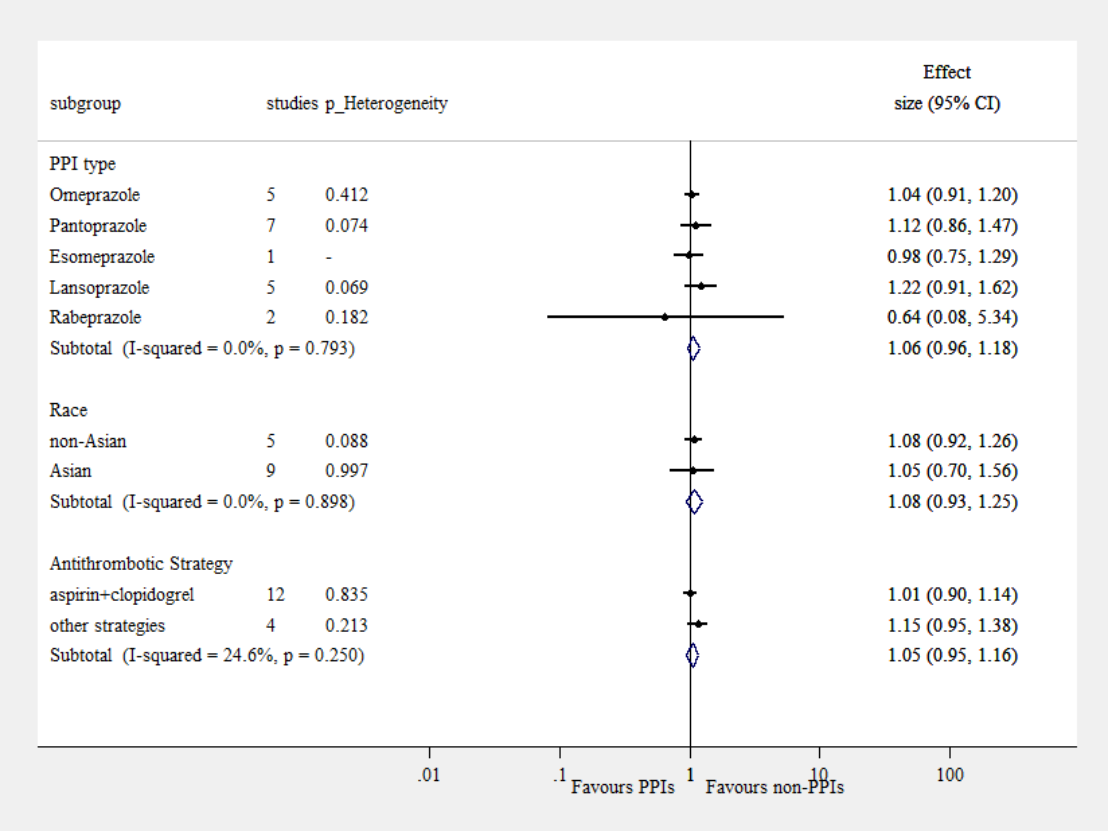
Supplementary Figure S7. Effect of PPIs combined with antithrombotic drugs on the risk of significant bleeding from gastroduodenal lesions in patients with coronary artery diseases.

Notes: D+L, the DerSimonian & Laird random effects model; M-H, the Mantel-Haenszel fixed effect model; PPIs, proton pump inhibitors.



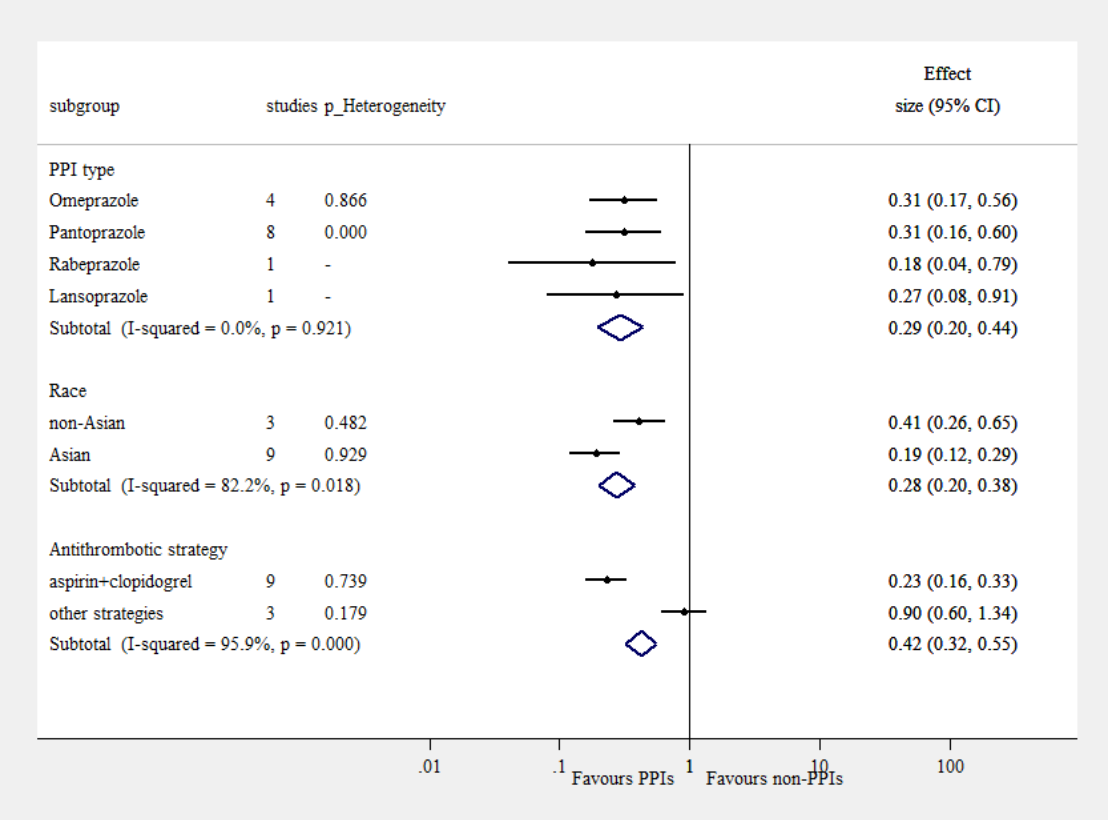
Supplementary Figure S8. Effect of PPIs combined with antithrombotic drugs on the risk of gastroduodenal ulcer in patients with coronary artery diseases.

Notes: D+L, the DerSimonian & Laird random effects model; M-H, the Mantel-Haenszel fixed effect model; PPIs, proton pump inhibitors.



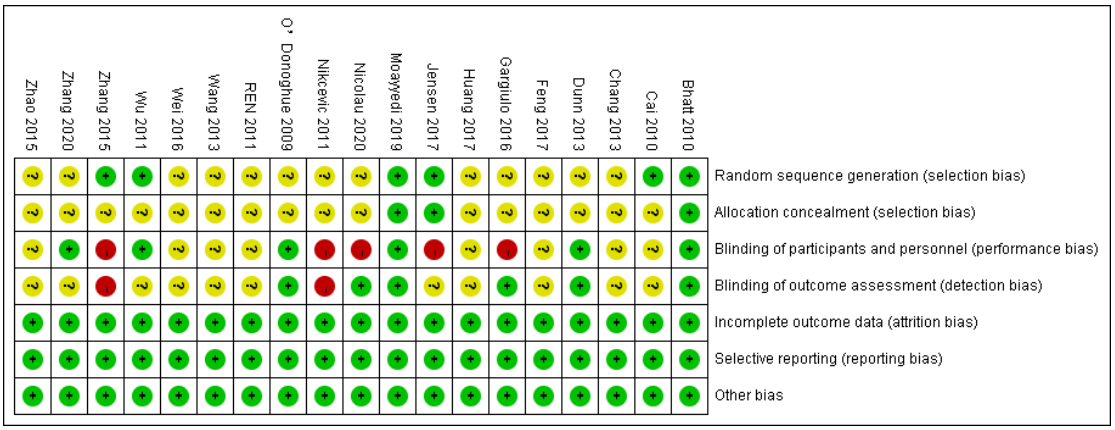
Supplementary Figure S9. Subgroup analysis of MACCE.

Notes: MACCE, major adverse cardiovascular and cerebrovascular events; CI, confidence interval; PPIs, proton pump inhibitors.

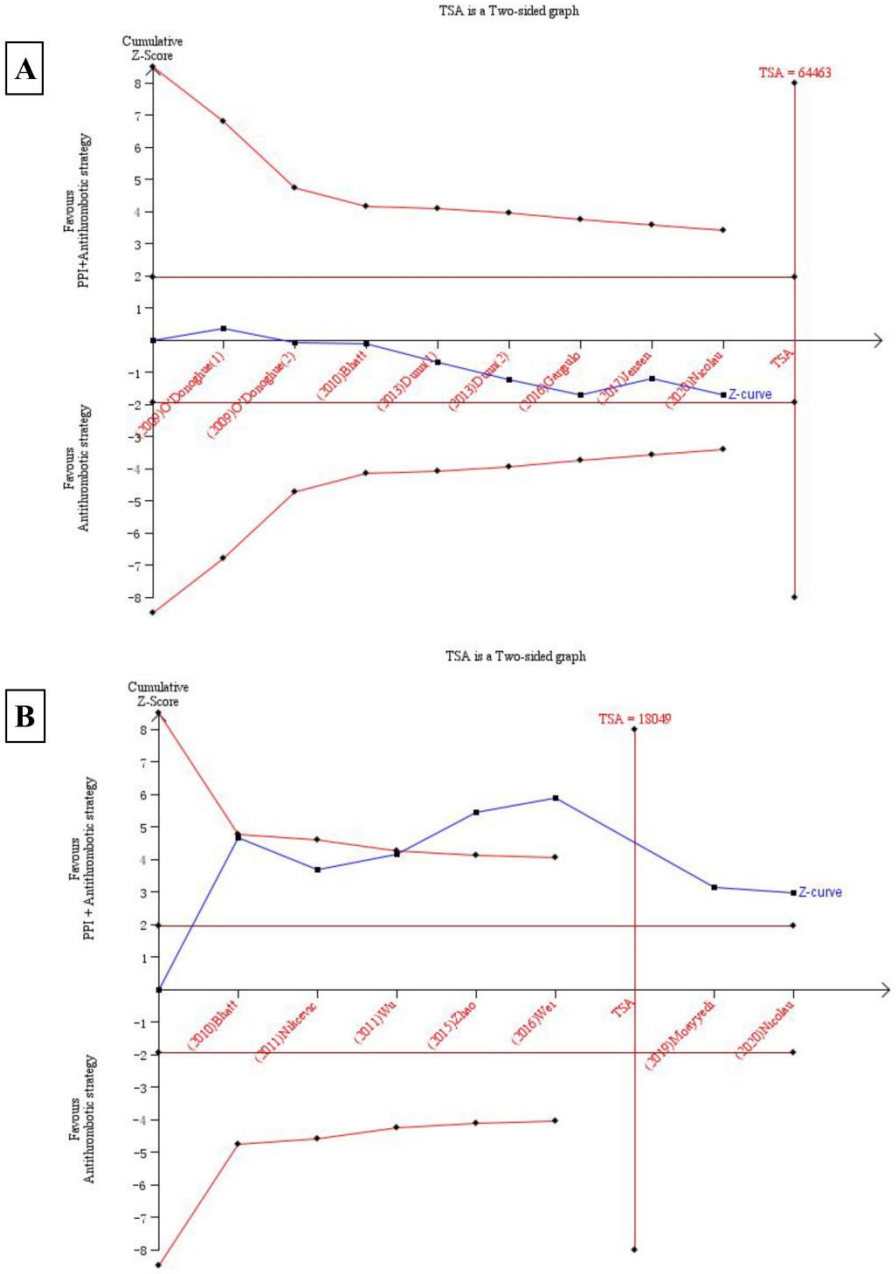


Supplementary Figure S10. Subgroup analysis of GI events.

Notes: GI, gastrointestinal; CI, confidence interval; PPIs, proton pump inhibitors.

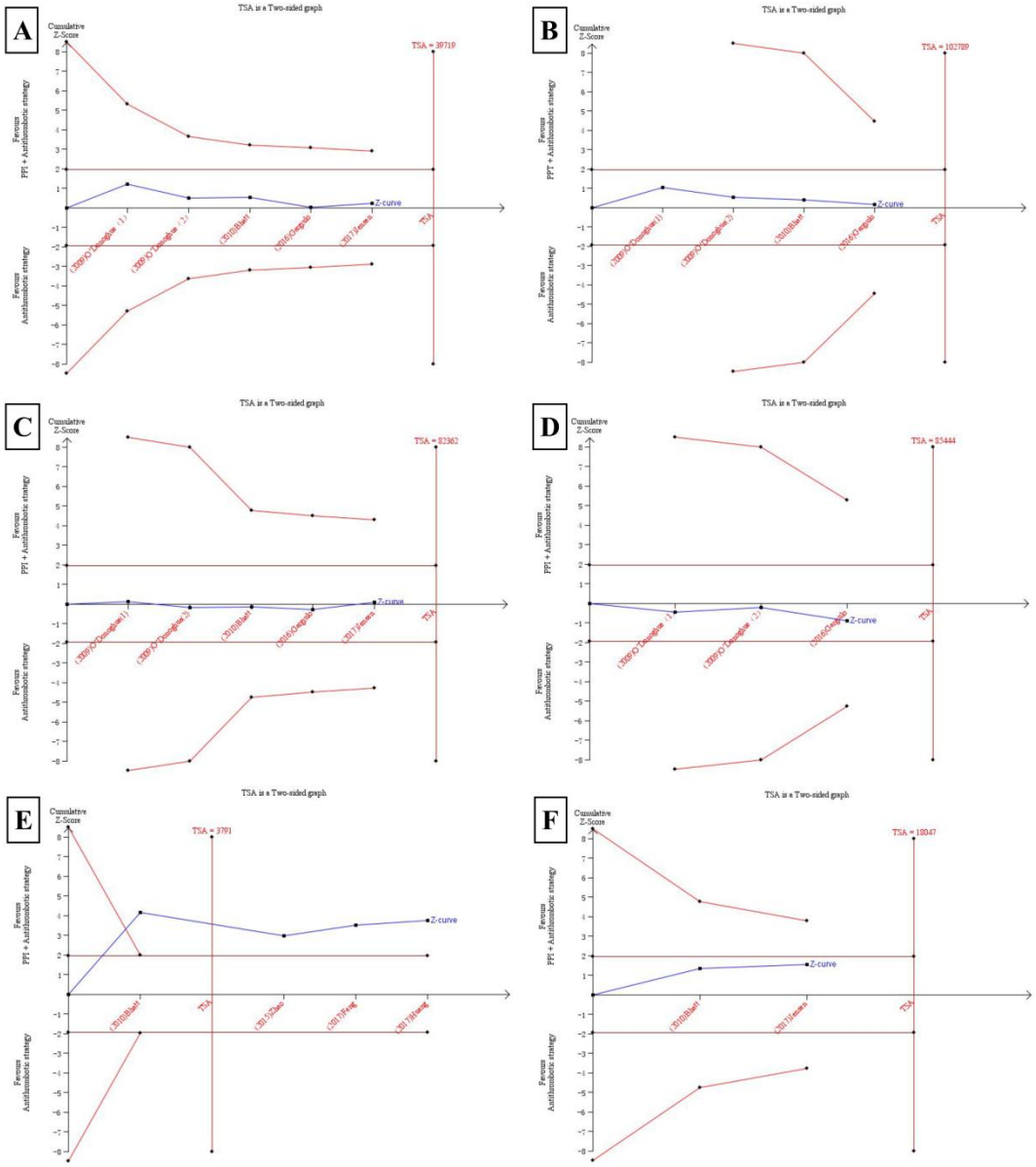


Supplementary Figure S11. Risk of bias assessments of each randomized controlled trial included.

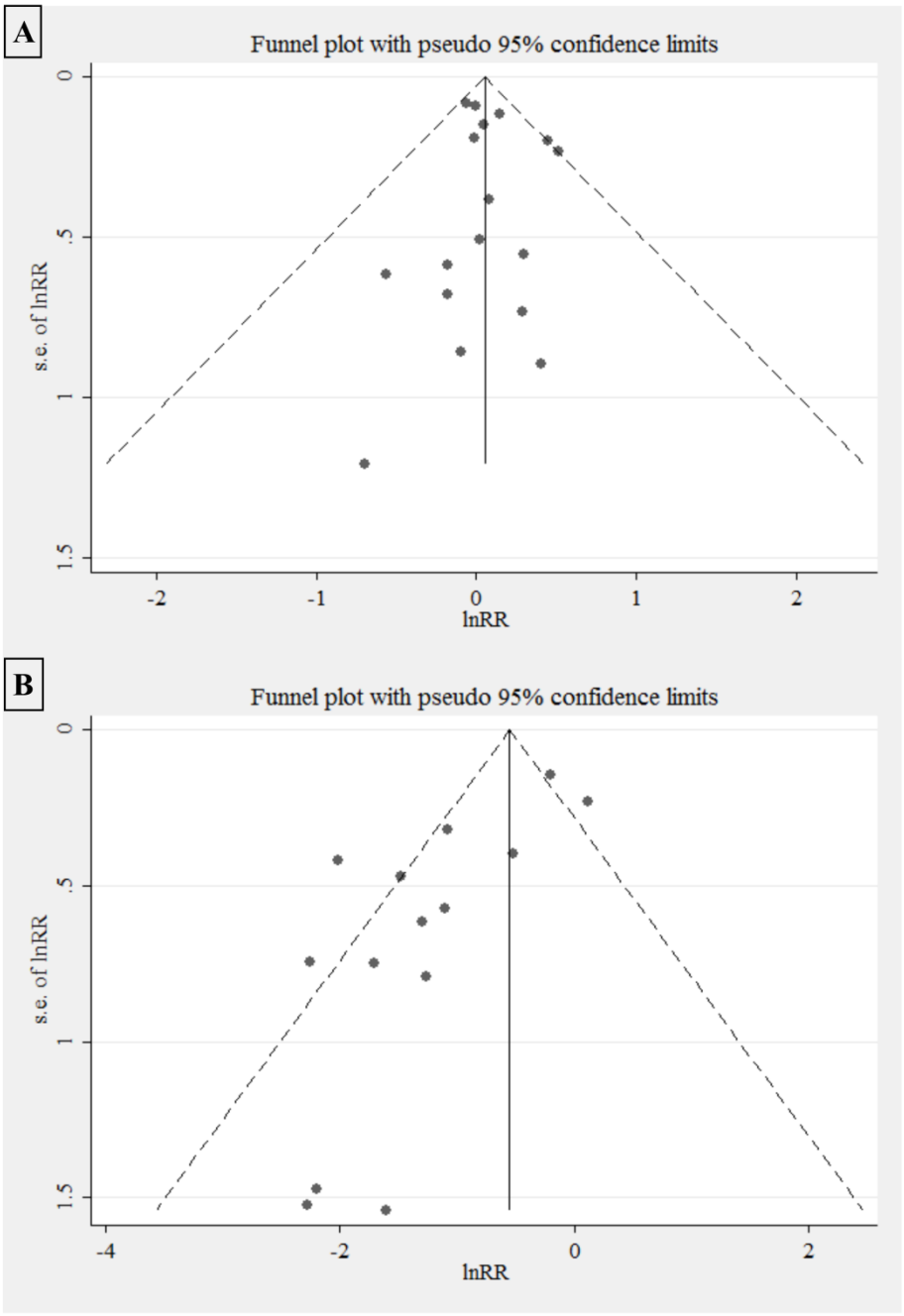


Supplementary Figure S12. Trial Sequential Analysis for MACCE (A) and GI events (B).

Notes: MACCE, major adverse cardiovascular and cerebrovascular events; GI, gastrointestinal.

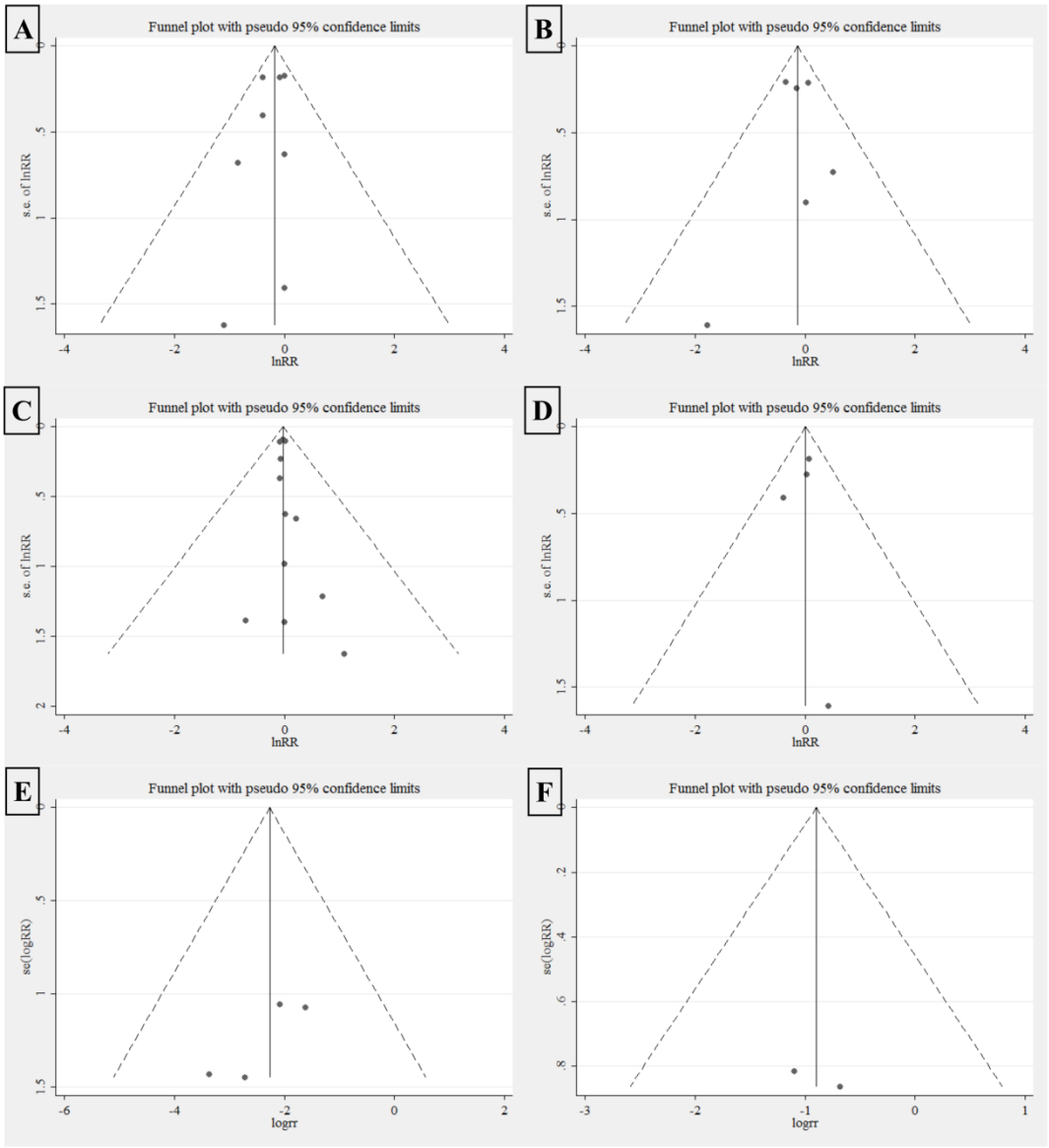


Supplementary Figure S13. Trial Sequential Analysis for all-cause death (A), cardiovascular death (B), myocardial infarction (C), stent thrombosis (D), significant bleeding from gastroduodenal lesions (E), and gastroduodenal ulcer (F).

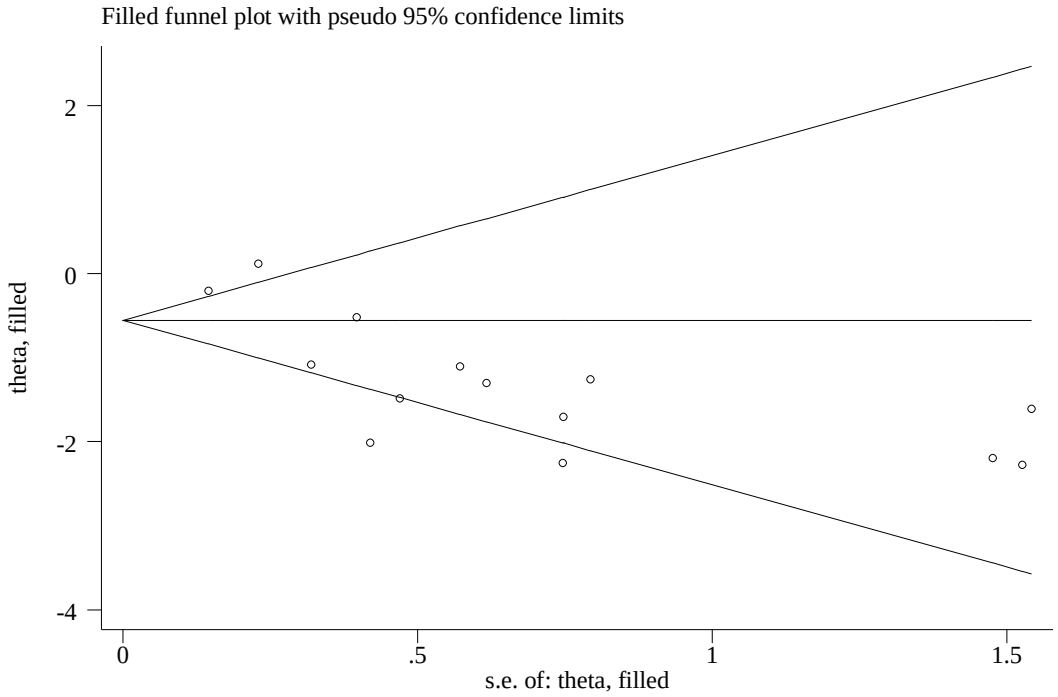


Supplementary Figure S14. Funnel plots of MACCE (A) and GI events (B).

Notes: MACCE, major adverse cardiovascular and cerebrovascular events; GI, gastrointestinal.



Supplementary Figure S15. Funnel plots of all-cause death (A), cardiovascular death (B), myocardial infarction (C), stent thrombosis (D), significant bleeding from gastroduodenal lesions (E), and gastroduodenal ulcer (F).



Supplementary Figure S16. The meta-trim operation for publication bias of GI events.

Notes: GI, gastrointestinal.